

discopt-speciation

Chemical equilibrium you can trust

discopt-speciation: aqueous speciation as an optimization plugin

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Outline

1. The problem: which phases are present?
2. Formulation: complementarity, not least squares
3. Solving it: homotopy, active sets, three methods
4. Global stability certificates
5. Gases and pressure: Peng–Robinson, log K(P)
6. Validation against PHREEQC
7. Beyond forward solves: sensitivities, fitting, design
8. Using it: Python, YAML, web app, Claude skill

The problem

Given water, dissolved elements, candidate minerals and gases:

- Which species form, and how much?
- Which solids precipitate, which stay dissolved?
- How does that change with T, pH, pe, dose?

> The hard part is the **phase assemblage**: a discrete choice hidden inside a continuous equilibrium problem.

The equilibrium conditions

condition	equation
mass action	$\sum_i \nu_{ri} \log_{10} a_i = \log_{10} K_r(T, P)$
conservation	$A\, n = b$, rows of A span the left null space of ν
complementarity	$N_p \geq 0, \quad \mathrm{SI}_p \leq 0, \quad N_p\, \mathrm{SI}_p = 0$
activity	$a_i = \gamma_i(m, T)\, m_i$

Conservation laws are **computed** from the network: element, charge and site balances come for free.

Why complementarity?

Common shortcuts: assume every candidate solid is present, or minimize $\sum \mathrm{SI}^2$.

Their answers change when you:

- multiply a reaction by two
- add an inert polymorph
- list a phase that should have precipitated

> Complementarity = the KKT conditions of Gibbs energy minimization. The answer cannot depend on how reactions are written.

Invariance, demonstrated

Counterexamples from Laliwala et al. (2025):

case	change	$\sum \mathrm{SI}^2$ objective	discopt-speciation
CE2	baseline	1.05	NL(s) = 9.537 mmol
CE3	add inert MX′(s)	17.24	identical state
CE4	double MX(s) reaction	4.19	identical state

Relative change in the aqueous state across CE2/3/4: **exactly 0**.

Solving: three methods, one answer

`complementarity` (default)

1. speciate with no phases
2. smoothed Fischer–Burmeister homotopy
3. active-set polish

`gibbs`: minimize G directly

`enumerate`: solve every assemblage (≤ 12 phases), proves uniqueness

> On the Laliwala et al. cases all three agree: molalities to 2e-10, solids to 3e-11 relative.

When local is not enough

A **non-ideal solid solution** makes G non-convex.

Inside a miscibility gap, one homogeneous solid satisfies every local condition, but two solids have lower G.

Tangent plane distance of a phase at composition x:

$$D(x) = \sum_j x_j \left(\ln x_j - \ln 10\,\mathrm{SI}_j\right) + \frac{G_{ex}(x)}{RT}$$

$D(x) \geq 0$ for every phase and every $x \Rightarrow$ global minimum.

A stability certificate

- Pure and ideal phases: exact, $\min D = \ln 10 \; s$
- Non-ideal solid solutions: **interval branch and bound** over the simplex
- Unstable? Add a phase copy at the minimizer, split by lever rule, re-solve

```
stability: certified (tol 1e-06; 1 phase-splitting rounds)
  SS  present  min D in [-4.5e-07, -1.7e-16]  stable
```

PHREEQC ex10 aragonite–strontianite gap: both binodal compositions reproduced to **six digits**.

Physics included

Activity models

- ideal, Davies, Debye–Hückel
- Truesdell-Jones, B-dot, LLNL B-dot
- SIT, Pitzer (E-θ, MacInnes)

T and P: van 't Hoff or PHREEQC analytic; above 1 atm, species molar volumes

Phases and interfaces

- pure solids, solid solutions
- gases at fixed P or V, ideal or Peng–Robinson
- Gaines–Thomas ion exchange
- diffuse double layer surfaces

Databases: phreeqc, pitzer, wateq4f, llnl, sit (ThermoChimie)

Quick start

```
from discopt.speciation import PhreeqcDatabase

db = PhreeqcDatabase.load("phreeqc.dat")
s = db.system(["Ca", "C(4)"], phases=["Calcite", "Aragonite", "CO2(g)"])
r = s.equilibrate({"Calcite": 1.0}, fixed={"CO2(g)": -3.5})
print(r.present, round(r.pH, 3))        # ['Calcite'] 8.279
```

Aragonite is listed, but correctly absent: it is metastable.

Want it anyway? `equilibrate(..., suppress=["Calcite"])`.

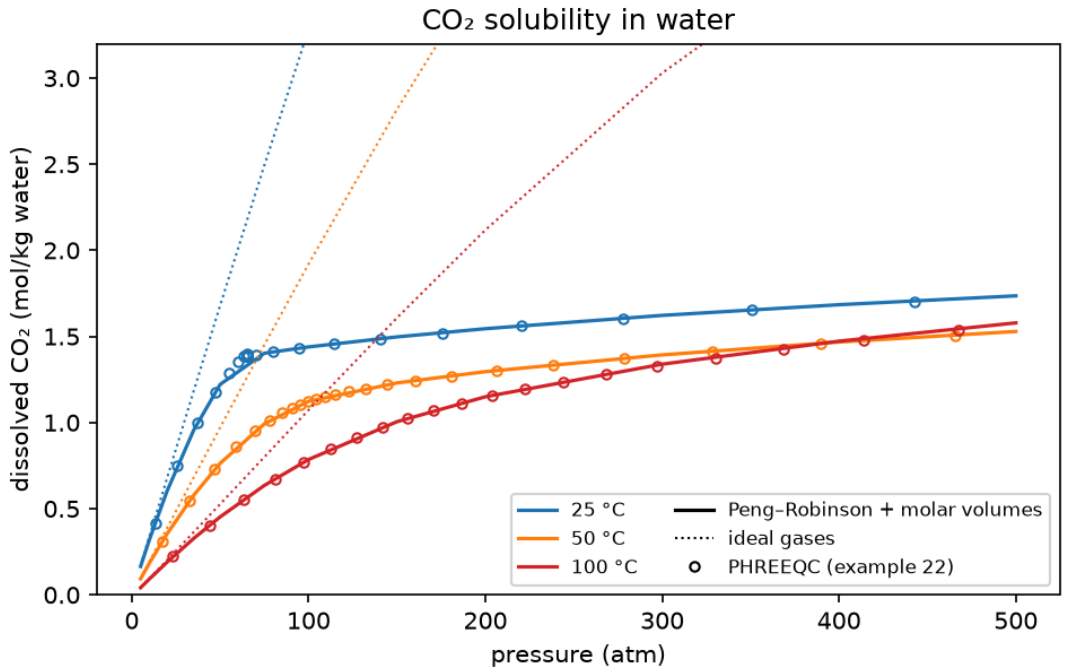
Or declare your own chemistry

```
from discopt.speciation import System

s = System("gypsum", activity="davies")
Ca, SO4, CaSO4 = s.aqueous("Ca+2"), s.aqueous("SO4-2"), s.aqueous("CaSO4")
s.reaction(Ca + SO4 >> CaSO4, logK=2.30, dH=5500.0)
s.reaction(s.solid("Gypsum") >> Ca + SO4, logK=-4.58, dH=-460.0,
           name="Gypsum")
print(s.equilibrate({"Gypsum": 1.0}, T=313.15).summary())
```

Also: strict YAML/JSON specs, and overrides on top of a database.

Gases at high pressure



- **Peng–Robinson** fugacities, PHREEQC's constants
- **log K(P)** from species molar volumes
- Pressure from the gas phase, as PHREEQC
- Ideal gases overestimate CO_2 several-fold

Validation against PHREEQC 3

case	model	agreement
NaCl + Ca, SO_4, 25/60 °C	Truesdell-Jones	log γ ≤ 6e-6
NaCl 1–4 m + Ca, SO_4	Pitzer	log a ≤ 5e-5
Th + CO_2, 25–150 °C	LLNL B-dot / SIT	log a ≤ 1.8e-4
Calcite + CO_2(g)	phreeqc.dat	pH 1e-7
Zn on Hfo, DDL	phreeqc.dat	ψ ≤ 1.4e-4 rel
Brine + calcite, gypsum, 100–500 atm	phreeqc.dat / Pitzer	pH ≤ 5e-7 / 2.4e-5
CO_2 + H_2O(g) to 1461 atm (ex22)	Peng–Robinson	pH ≤ 1.2e-6

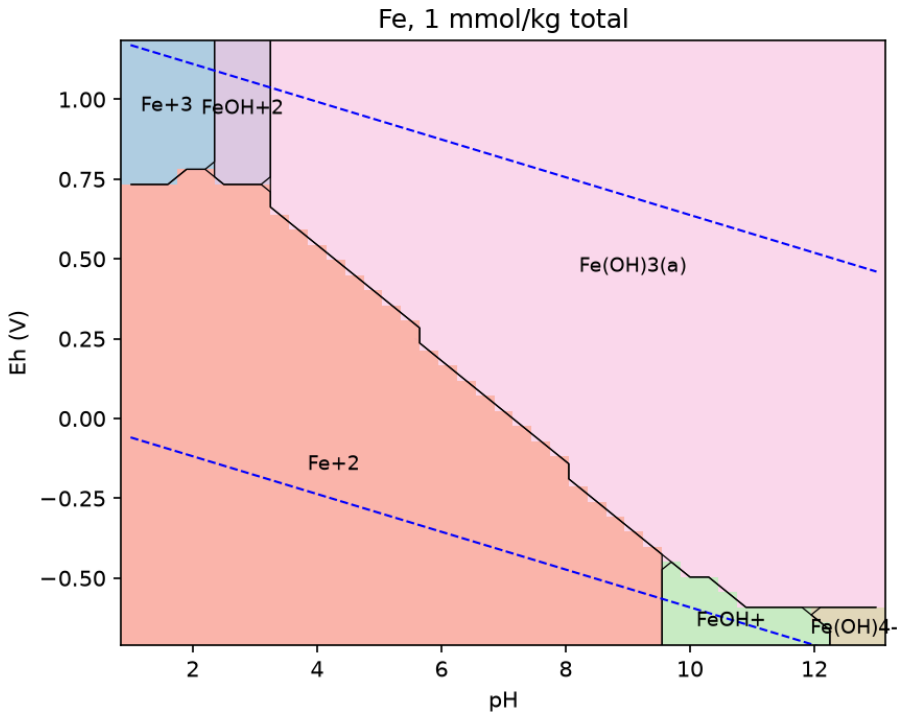
377 steps of the PHREEQC example problems and stress cases, each a test. Known limits are `xfail(strict=True)` with a diagnosis.

Honest about the limits

- **Log-molality floor** (−200): vanishing redox states can stall
- **Gas condensation**: not a separate phase (PHREEQC caps P in the loop)
- **Water gfw**: 18.016 vs 18.01528, a convention

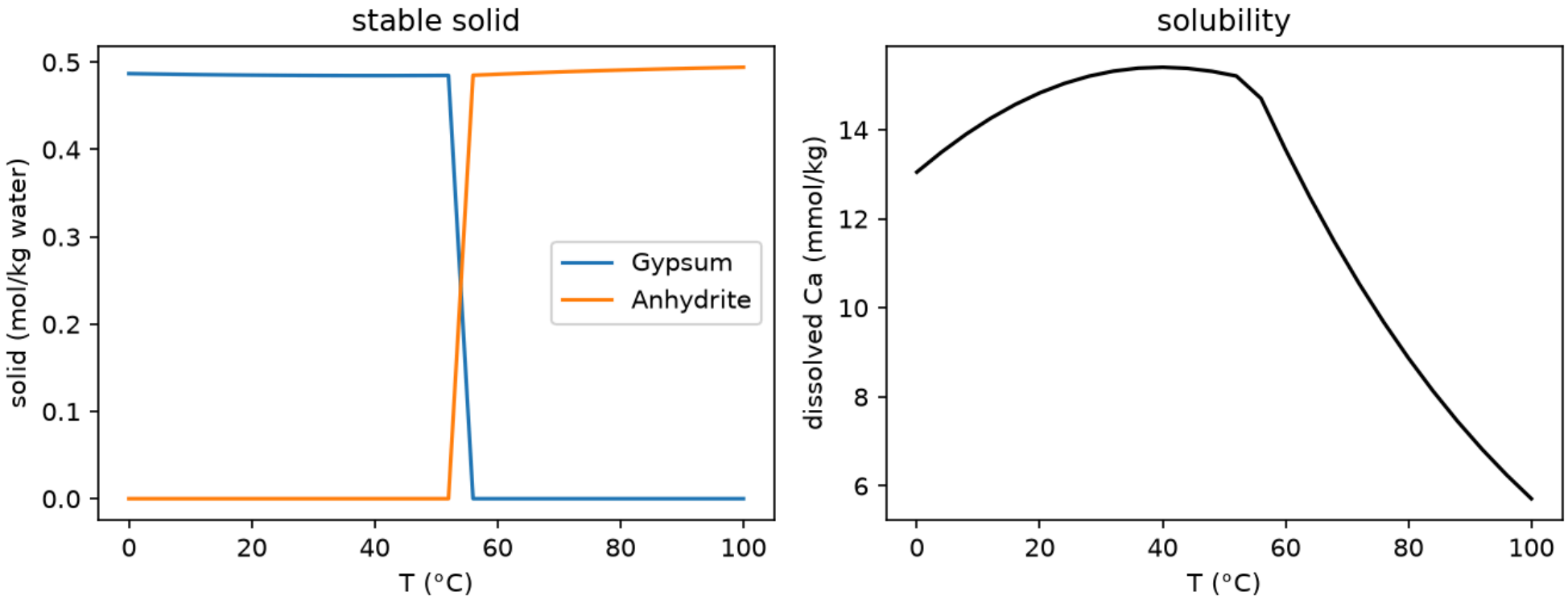
> Every result carries independent `residuals`. `converged` means all are below 1e-8.

Diagrams from full equilibria



- Sweeps, predominance, Pourbaix, Ellingham
- Every point is a full equilibrium
- Phase boundaries emerge from complementarity, not hand drawing

Gypsum to anhydrite, for free



The phase transition near 55 °C and the solubility kink come straight out of the T sweep.

Exact sensitivities

At a converged state the assemblage equations are square: $F(x, \theta) = 0$.

$$\frac{dx}{d\theta} = -F_x^{-1} F_\theta$$

- F_x exact via JAX forward mode
- No extra solves
- Phases on a kink ($N_p = 0, \mathrm{SI}_p = 0$): one-sided derivatives, reported as such

Fitting constants to data

Calcite at 8 CO_2 pressures; measure pH and dissolved Ca (synthetic, noise added).

```
chi2 = 15.07, dof = 14, reduced chi2 = 1.077
parameter            value    std err   95% CI
logK Calcite       -8.4711    0.0095   [-8.4897, -8.4525]
logK CaHCO3+        11.471      0.15   [11.178, 11.764]
```

True values −8.480 and 11.435: both inside the intervals. Profile likelihood available too.

Design and worst-case bounds

Equilibrium becomes a constraint inside a discopt model.

```
r = O.design(s, {"Al": 1e-3, "Cl": 3e-3}, {"NaOH": (0, 6e-3)},
             objective="NaOH",
             constraints=[(O.dissolved("Al"), "<=", 1e-6)],
             reagents={"NaOH": {"Na": 1, "OH-": 1}})
# ('local_optimum', {'NaOH': 0.0029818}, True)
```

`worst_case_bounds`: min/max of a quantity over a box of log K uncertainty, via spatial branch and bound.

Certified only when every bound is rigorous.

Try it in the browser

- Pyodide + the **pounce** NLP solver in WebAssembly
- Same algorithm as the package, nothing on a server
- Speciation, sweeps, Pourbaix, log K(T), exported Python
- The URL holds the setup: share a link, share a calculation

kitchingroup.cheme.cmu.edu/discopt-speciation/app/

Ask in plain language

```
discopt-speciation-install-skill
```

- `/discopt-speciation` Claude Code skill
- Five tested example scripts to copy from
- A `speciation-expert` agent for difficult cases

"What dose of NaOH keeps dissolved Al below 1 µM?"

Summary

- Complementarity gives answers independent of how chemistry is written
- Interval certificates prove global stability with solid solutions
- Non-ideal gases and pressure-dependent log K, to 1500 atm
- Validated against PHREEQC across 377 example steps
- Sensitivities, fitting with uncertainty, and design come built in
- Python, YAML, browser, or a conversation

Thank you

github.com/jkitchin/discopt-speciation

kitchingroup.cheme.cmu.edu/discopt-speciation

Parkhurst & Appelo (2013) PHREEQC 3, doi:10.3133/tm6A43
Laliwala et al. (2025) Syst. Control Trans. 4, 2140, doi:10.69997/sct.132742
Michelsen (1982) Fluid Phase Equilib. 9, 1, doi:10.1016/0378-3812(82)85001-2